

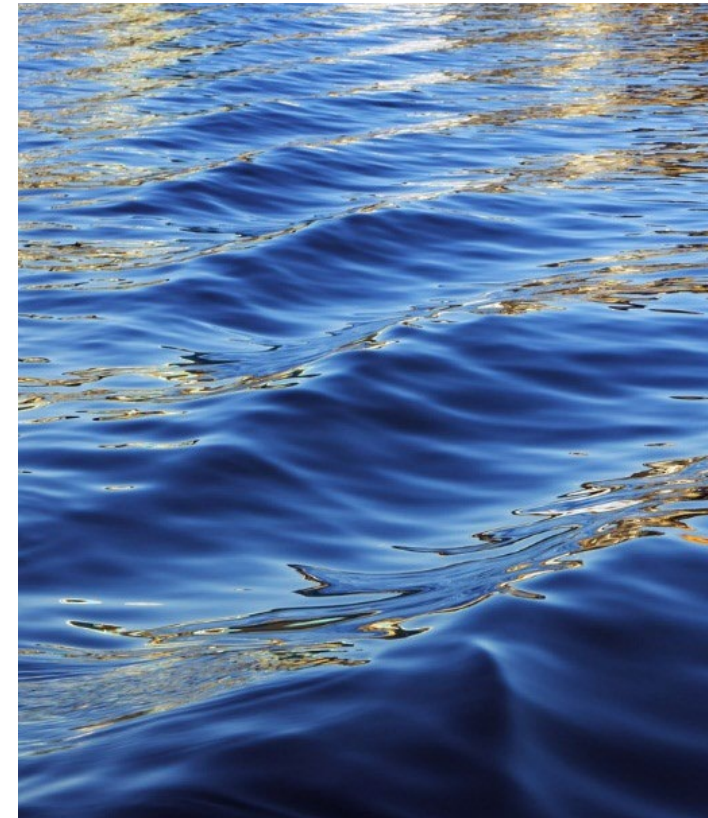


Digital Twin Practice

Demand and operations variability



European University Alliance for Global Health



Co-funded by the
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Universidad
de Alcalá



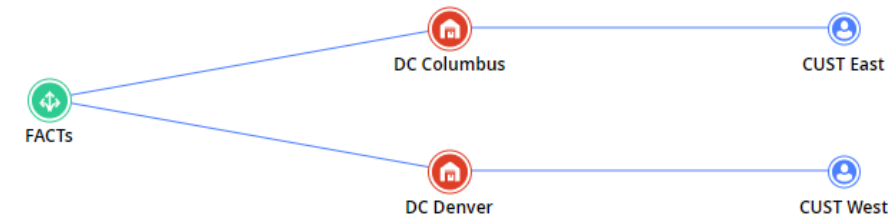
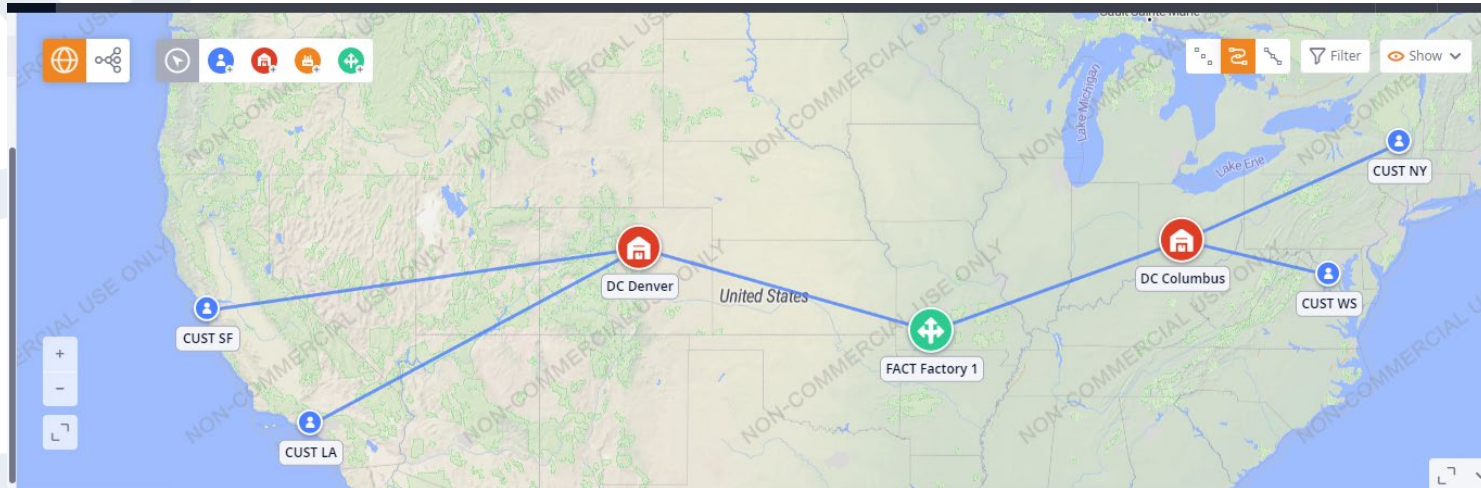
Learning objectives

- Learn to simulate with variability:
 - Variable demands
 - Changing operations
- Quantitative Analysis
- Qualitative Analysis ← Use an AI for this!



The Use Case

- ACME Inc. is producing plastic components from its factory in Springfield, Illinois, USA.
- It handles two distribution centers to serve USA market:
 - DC1 in Columbus, to serve East Coast Customers located in New York and Washington.
 - DC2 in Denver, to serve West Coast Customers located in San Francisco and Los Angeles.





The Use Case (II)

- The product is measured in pallets (uds):
 - Unit cost of 1,5K\$ and a selling price of 2K\$.
 - Pallet size is 1m³.
- Demand:
 - Periodic demand, with next day after interval, 1 day interval, amount of 200 uds.
- Inventory policies:
 - Min/Max with safety stock : $s=5$, $S=10$, $ss=2$
 - Initial stocks: 3.000 uds at factories and distribution centers.
 - Facility costs: 12 \$/pallet/day.
- Transport policies:
 - By road, in trucks with a capacity of 20m³, and a mean speed of 50km/h.
 - Cost based on product*distance: 0,05cts/pallet/km.
 - Transport time: 5 days between factory and DCs , and 3 days between DCs and customers.
- Shipping policy is LTL, based on demand.



Prepare the dashboard for our analysis

- Use “simulation experiments” (we are not designing).
- Dashboard to monitor activity:

Area	Purpose	KPIs
Fulfillment	How much demand are we receiving? Are we managing to meet it on time?	Demand Placed, Fullfillment received, Fullfillment received by customers, Fulfillment received on time.
Available Inventory	See how the inventory evolves in the factory and DCs.	Available Inventory, Daily Available inventory
Finantials	What is the profit generated from sales revenue and costs incurred? What is the breakdown by cost type and customer?	Revenue, Total Cost, Transportation Cost, Facility Costs
Lead Time	How much time takes to deliver the orders to the customers ?	Lead Time over time.
Service Level	Are we fulfilling our promise of on-time delivery to our customers?	ELT per product.
Peak Capacity	What is our peak capacity at our premises? Are we able to serve everything?	Peak Capacity per site

Scenario 1: the baseline

Please edit selected cell(s) ✕

First occurrence

Next day after interval ▼

Order interval, days

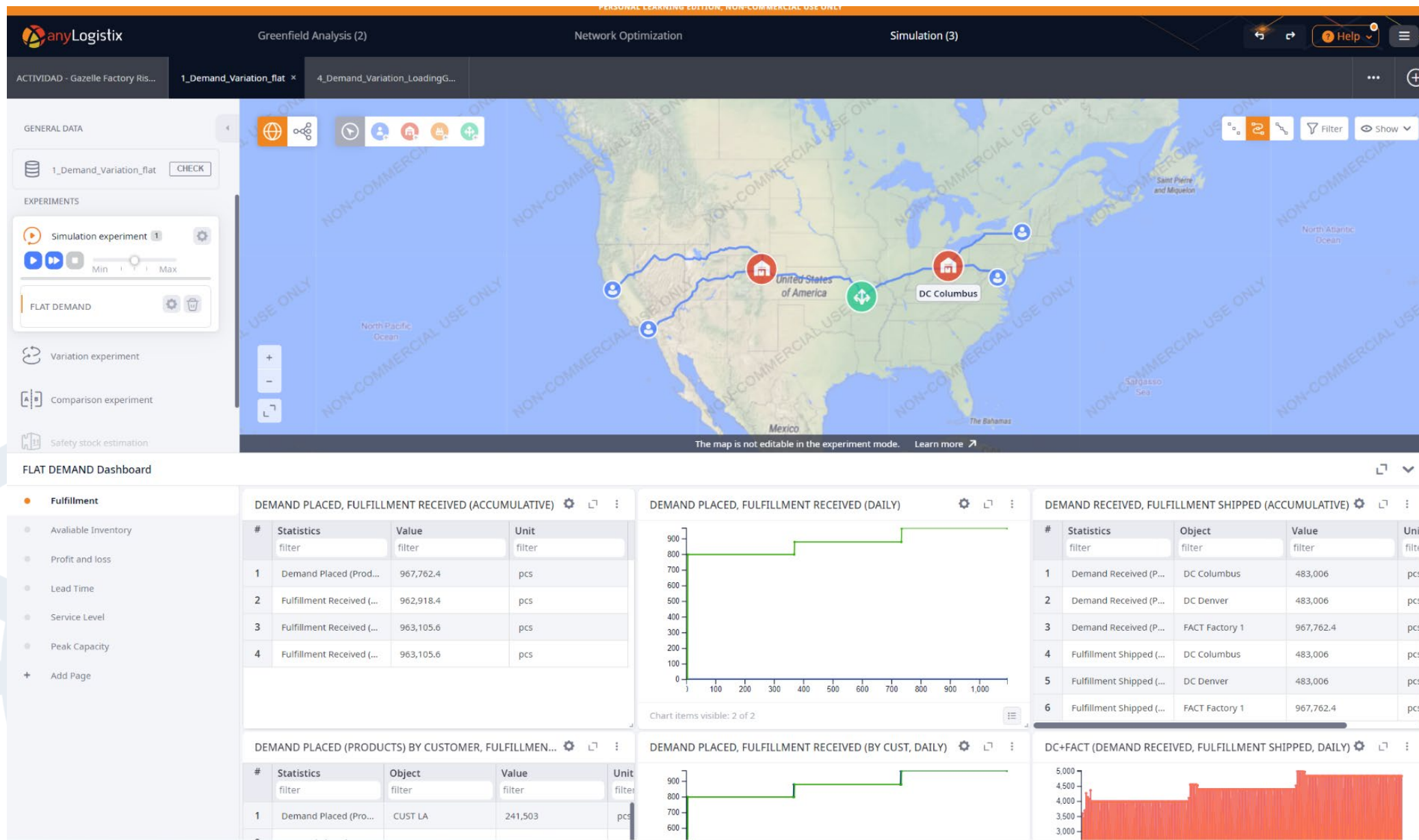
1 📊

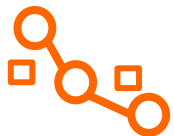
Quantity

Normal(50; 200) 📊

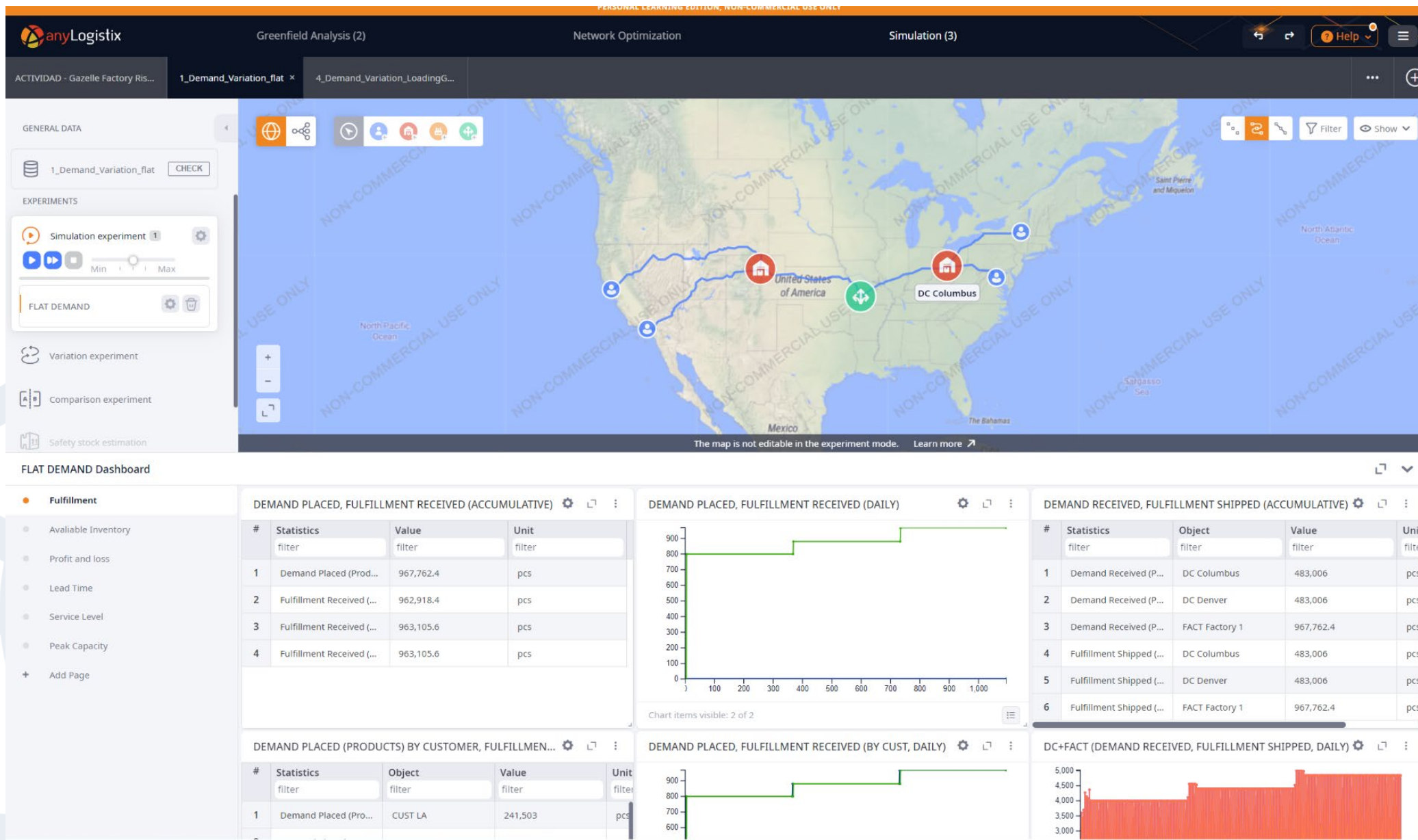
Cancel Save

#	Customer	Product	Demand Type	Parameters
1	[CUSTs]	Pallet	Periodic demand	Order interval: 1, Qu...





Fulfillment

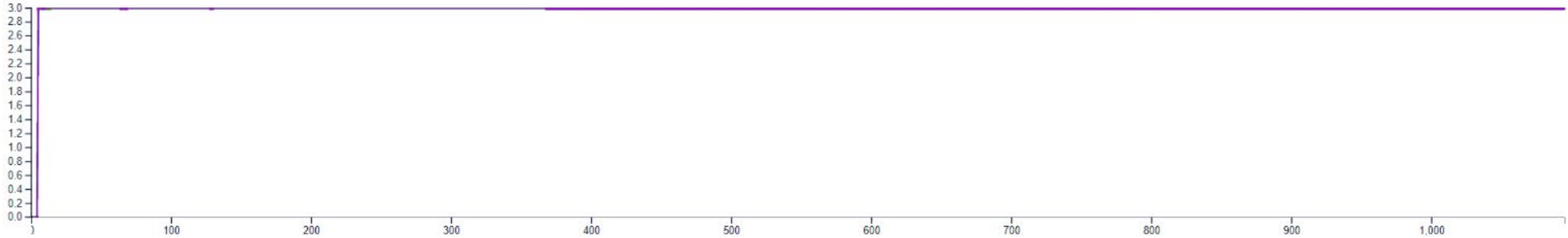




Service Levels



LEAD TIME



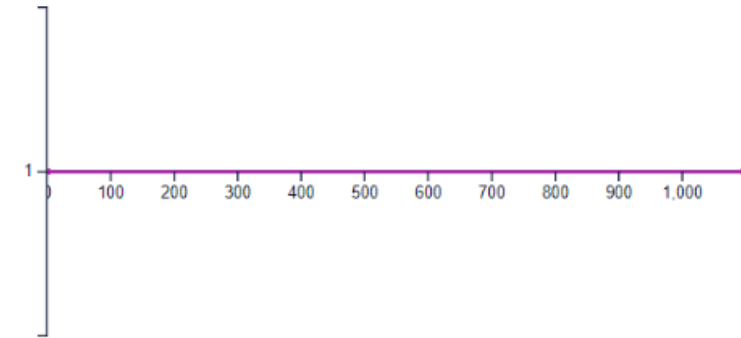
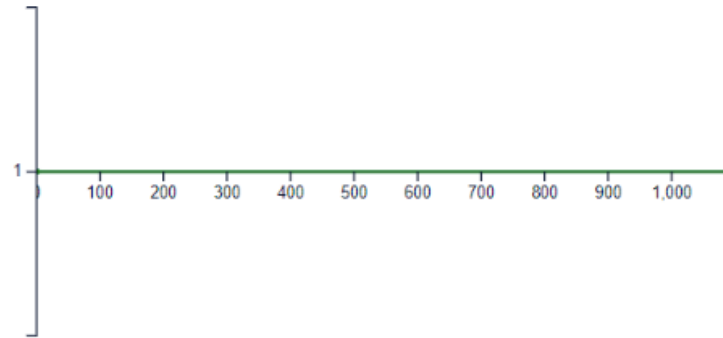
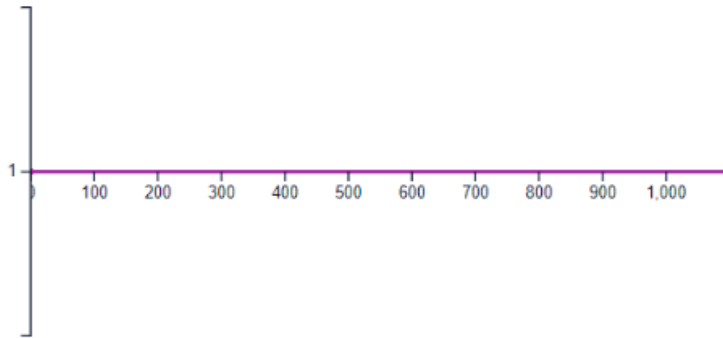
ELT SERVICE LEVEL BY PRODUCTS (PER PRODUCT)



ELT SERVICE LEVEL BY PRODUCTS (PER SOURCE)



ELT SERVICE LEVEL BY REVENUE (PER OBJECT)



Scenario 2: variable demand

Please edit selected cell(s) ×

First occurrence

Next day after interval ▼

Order interval, days

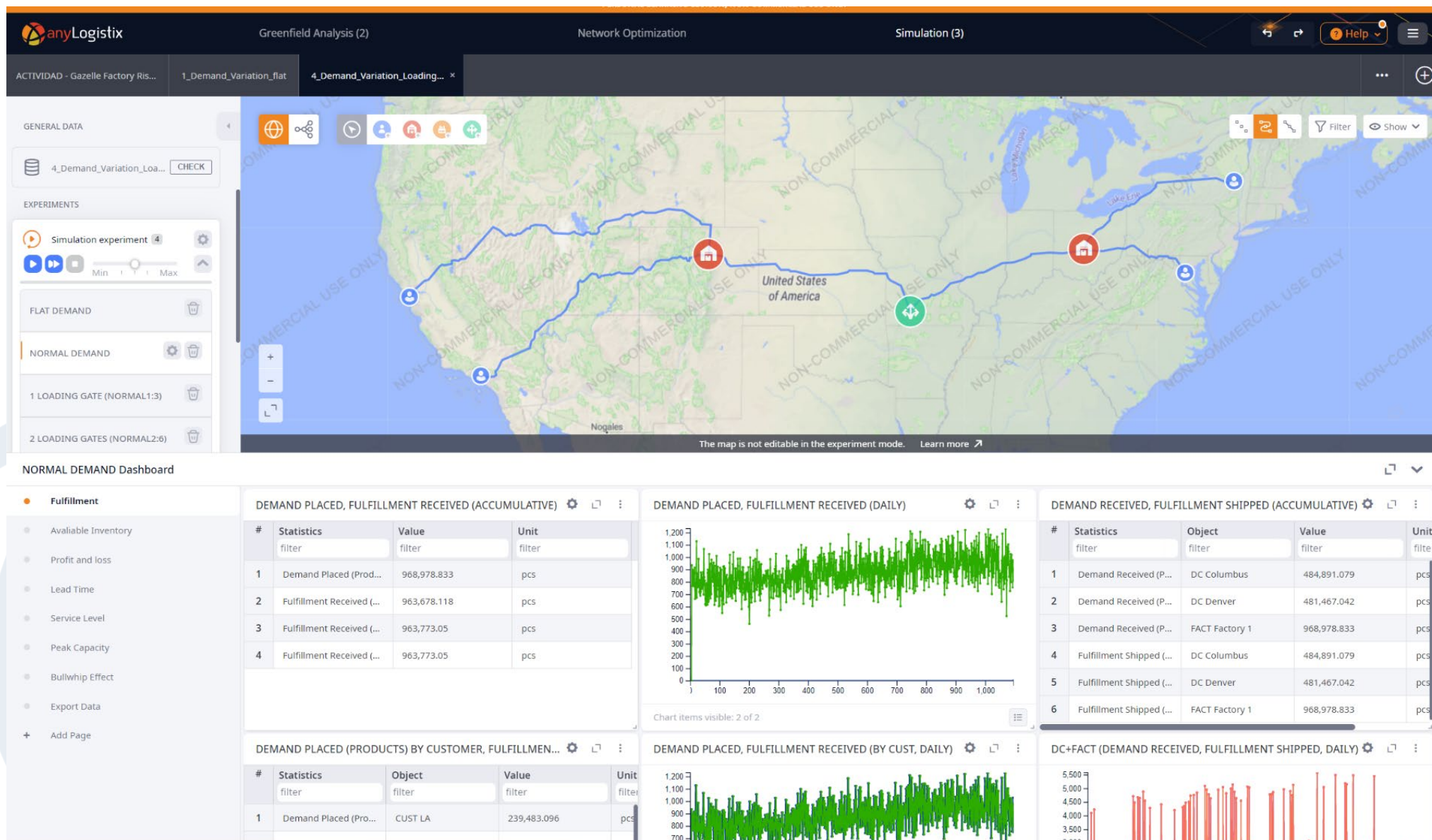
1 📊

Quantity

Normal(50; 200) 📊

Cancel Save

#	Customer	Product	Demand Type	Parameters
1	[CUSTs]	Pallet	Periodic demand	Order interval: 1, Qu...



Scenario 3: variability in operations

Basic All In Use (17)

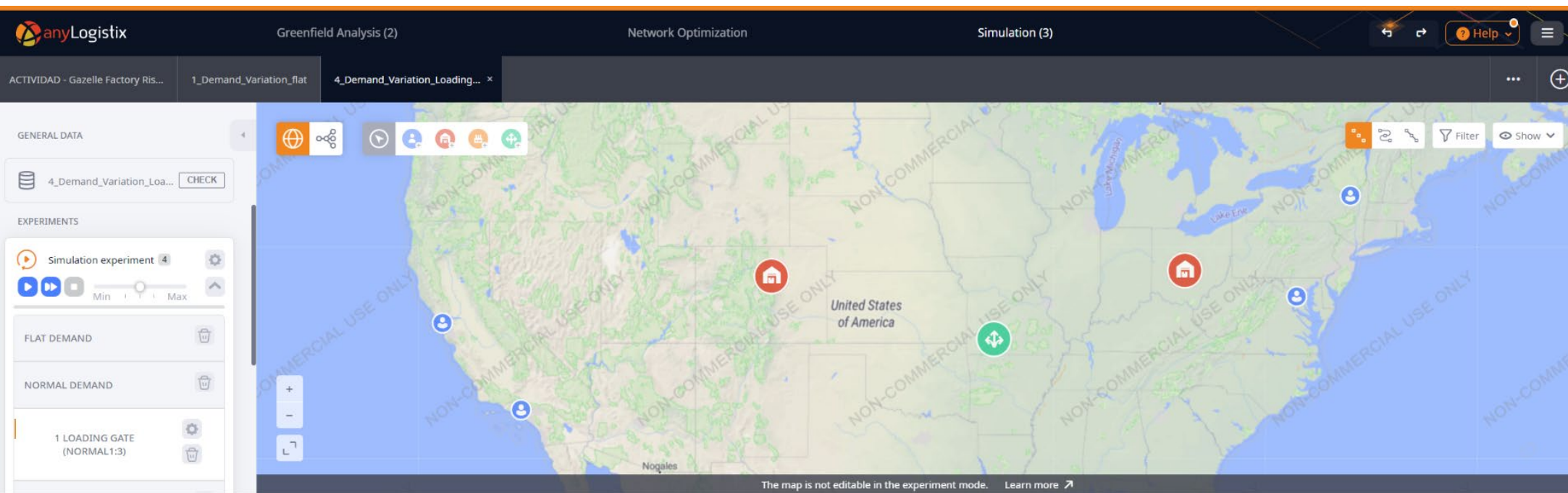
filter

Groups (5)

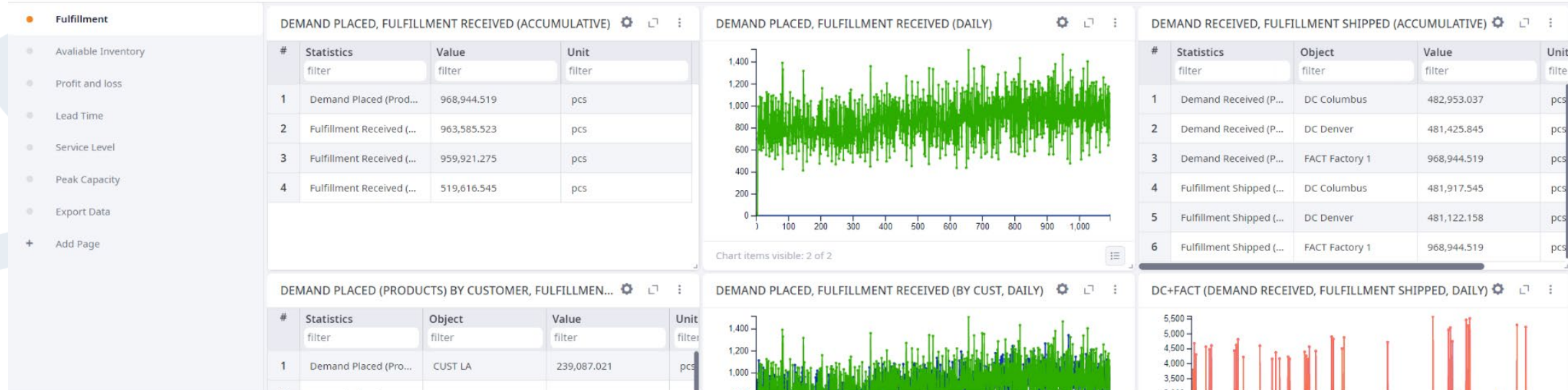
Inventory (2)

Loading and Unloading Gates (1)

#	Name	Facility	Type	Vehicle Type	Number of Gates	Units	Processing Time	Time
1	Facility Gates	[DCs]	Loading gate	Truck	1	m ³	Normal(1; 3)	min

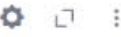


1 LOADING GATE (NORMAL1:3) Dashboard

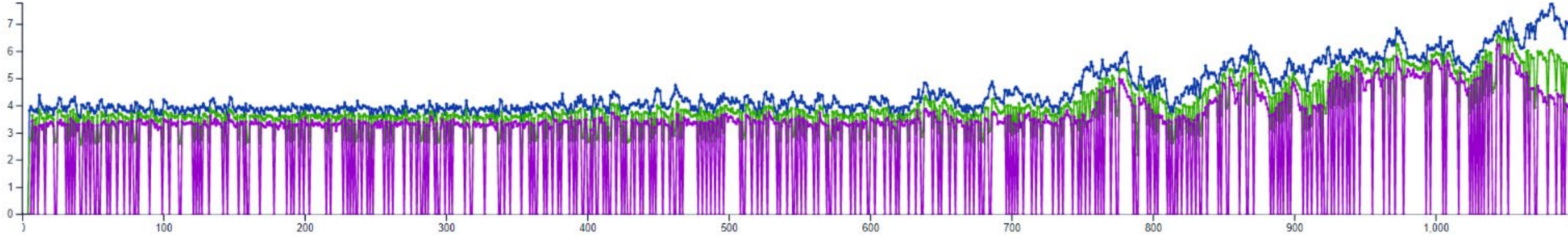




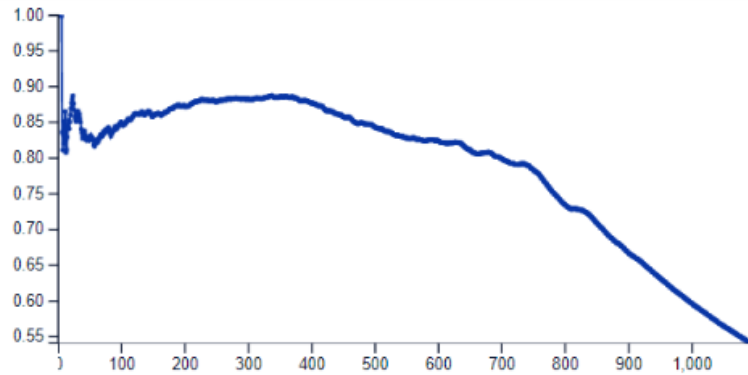
Service Level



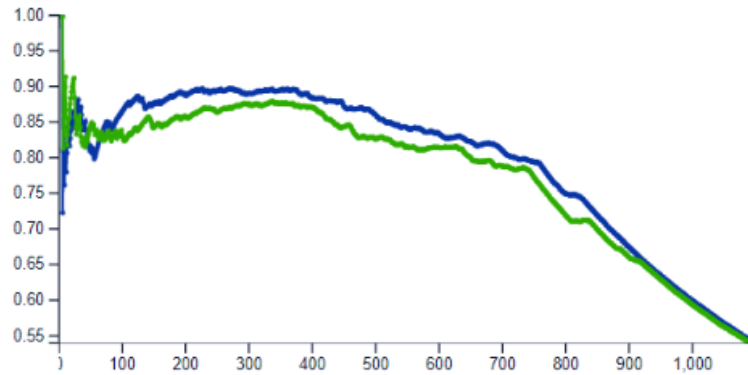
LEAD TIME



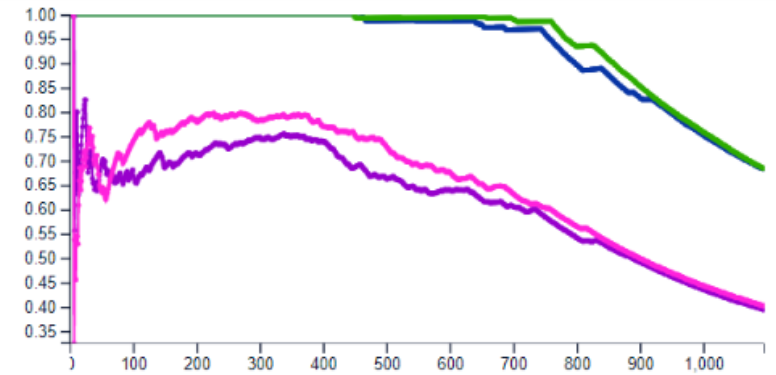
ELT SERVICE LEVEL BY PRODUCTS (PER PRODUCT)



ELT SERVICE LEVEL BY PRODUCTS (PER SOURCE)



ELT SERVICE LEVEL BY REVENUE (PER OBJECT)



Scenario 4: increasing variability

Basic All In Use (17)

filter

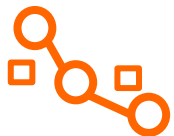
Facility Expenses (1)

Groups (5)

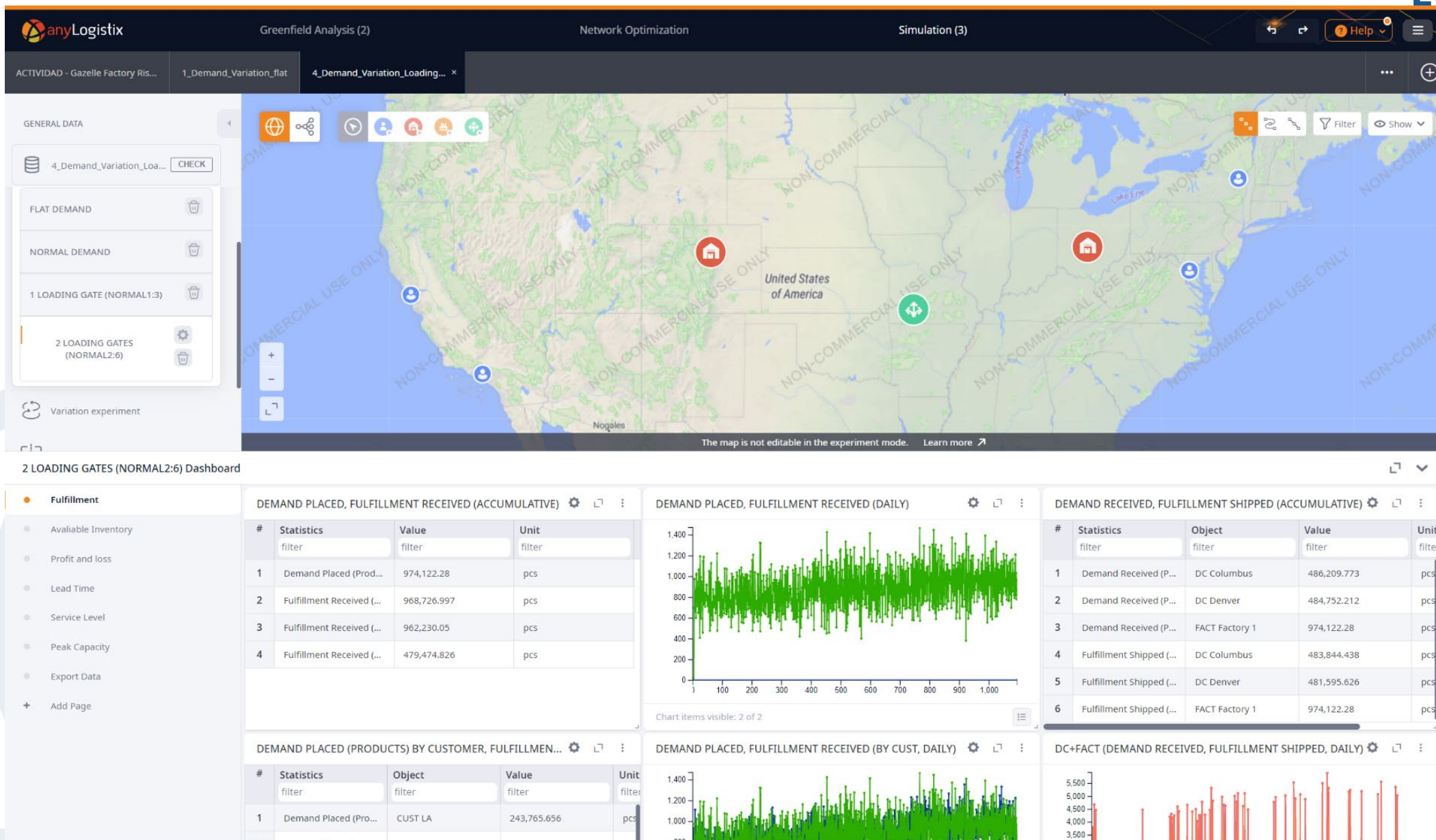
Inventory (2)

Loading and Unloading Gates (1)

#	Name	Facility	Type	Vehicle Type	Number of Gates	Units	Processing Time	Time U
1	Facility Gates	[DCs]	Loading gate	Truck	2	m ³	Normal(2; 6)	minut



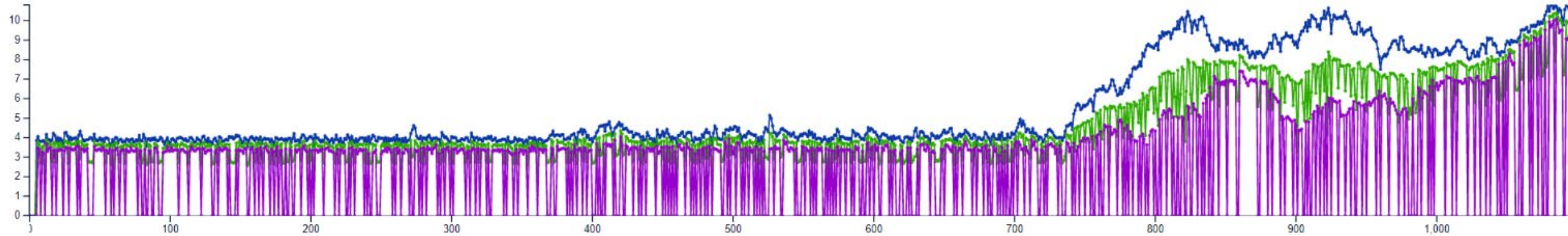
Fulfillment



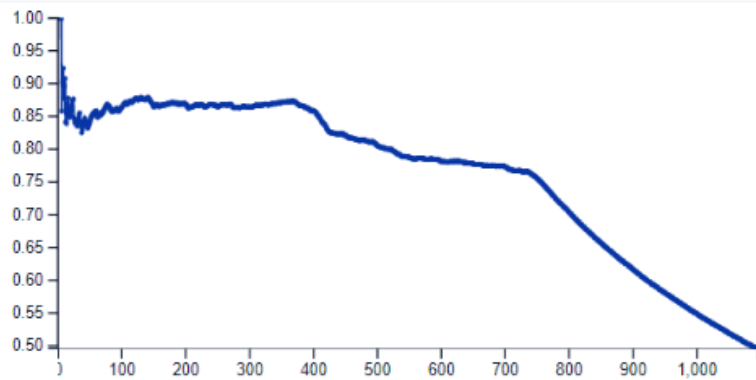


Service Level

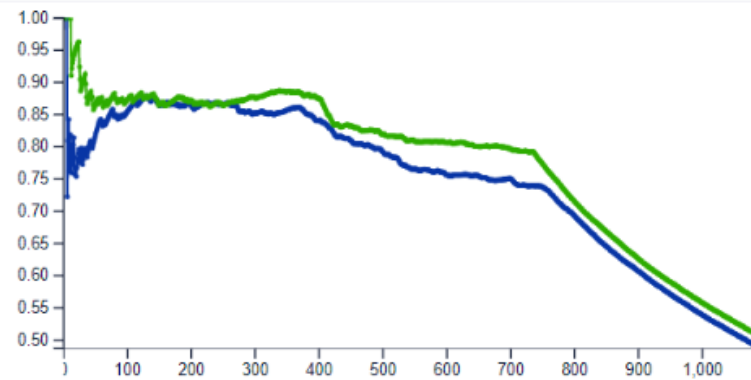
LEAD TIME



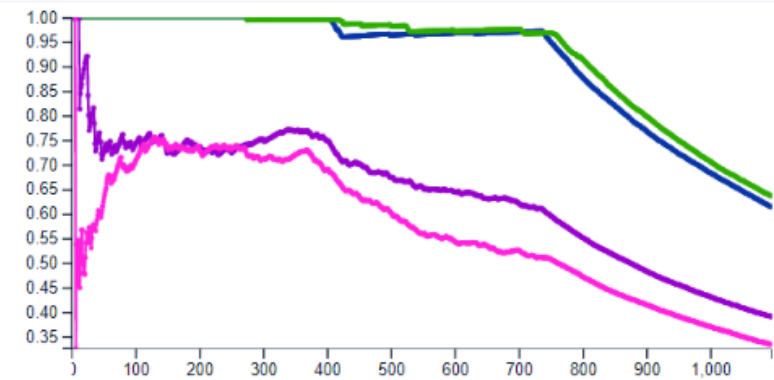
ELT SERVICE LEVEL BY PRODUCTS (PER PRODUCT)



ELT SERVICE LEVEL BY PRODUCTS (PER SOURCE)



ELT SERVICE LEVEL BY REVENUE (PER OBJECT)

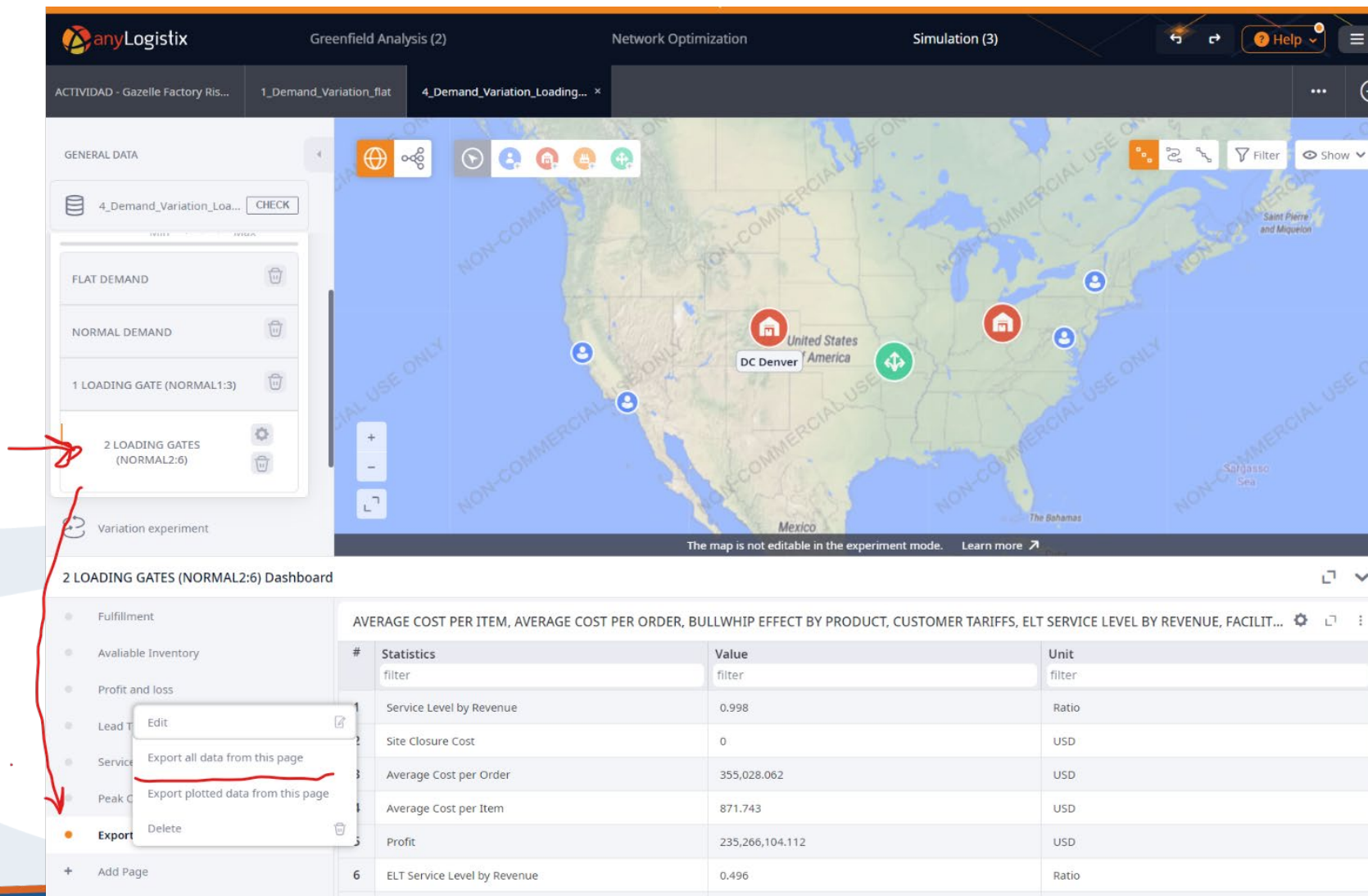


Use AI for Quantitative and Qualitative Analysis



How to compare all the scenarios?

- Export the experiment data and use an AI



The screenshot displays the anyLogistix software interface. The top navigation bar includes tabs for 'Greenfield Analysis (2)', 'Network Optimization', and 'Simulation (3)'. The main area is divided into a left sidebar and a right panel. The sidebar contains a 'GENERAL DATA' section with a 'CHECK' button and a list of demand and loading gate configurations. A red arrow points to the 'Export' button at the bottom of the sidebar. A red circle highlights the 'Export all data from this page' option in the dropdown menu. The right panel shows a map of the United States with several location markers. Below the map is a dashboard titled '2 LOADING GATES (NORMAL2:6) Dashboard' which includes a table of statistics.

#	Statistics	Value	Unit
1	Service Level by Revenue	0.998	Ratio
2	Site Closure Cost	0	USD
3	Average Cost per Order	355,028.062	USD
4	Average Cost per Item	871.743	USD
5	Profit	235,266,104.112	USD
6	ELT Service Level by Revenue	0.496	Ratio



The prompt

- In the role of "supply chain process analyst", I want to generate a quantitative comparative report (in English) for the result of 4 different simulations performed of a supply chain.
- Each of the simulations has dumped its results into an Excel sheet where the name of the simulation corresponds to the name of the Excel file without the extension ".xlsx".
- Each of the simulations collects its results within the "Table 1" sheet, in a data table from row 7 onwards, and columns A, B and C. Column A corresponds to the name of the indicator (KPI) obtained, column B corresponds to the value obtained for the KPI, and column C corresponds to the units of the KPI.
- For the analysis, you will perform the following steps:
 - 1. Create a comparison table where the rows of the column will correspond to the values obtained in column A, and as many columns as simulation scenarios are collected will be included, indicating in each column the VALUE of each KPI.
 - 2. Delete rows whose values are all "0".
 - 3. You will do a quantitative analysis for each simulation scenario including the assessment of all its KPIs, analyzing the pros and cons of each scenario. You will group the financial criteria first, then the service criteria and finally the rest of the criteria.
 - 4. Add a detailed root cause analysis to the report, justifying and explaining each of the variations. 5. You will upload the report as a professional-quality PDF file.
- the report title will be "Comparative Analysis of Supply Chain Simulation Scenarios" and include a summary page or executive overview at the beginning of the PDF



The report

